

VRm: A Virtual Reality Tool for Anatomical Study

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ABSTRACT

VRm is the prototype for a Virtual Reality (VR) software tool for medical professionals to study anatomy. The tool is designed to improve learning outcomes for complex anatomical subject matter that is difficult to teach using traditional methods. VRm was developed at the Rochester Institute of Technology by faculty and students. In addition to developing the tool, the goal of the project was to explore pedagogical approaches emphasizing cross-disciplinary collaboration, production-based learning, and to utilize industry-standard software development methodologies in the context of a class.

CCS CONCEPTS

• **Applied computing;** • **Education;** • **Collaborative learning;**
• **Media arts;** • **Life and medical sciences;**

KEYWORDS

VR, Undergraduate, Anatomy

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1 INTRODUCTION

There were two fundamental goals for this project, both of equal importance. First, we wanted to develop a pedagogical approach that was novel for RIT. Second, we wanted to develop a software tool to help medical professionals study anatomy using Virtual Reality.

The Rochester Institute of Technology (RIT) comprises nine colleges, each focused on different educational disciplines. Cross-disciplinary collaboration is always desired but can be challenging to implement due to practicalities such as lab space, computer access, curriculum differences, and scheduling complexities. Our goal was to work at the convergence of medical illustration and real-time rendering (game) technology, and to use game technology for a non-game project. To that end, we gathered faculty and students from the School of Interactive Games and Media (IGM), and from the Medical Illustration (MI) major, to collaborate on a single project.

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Those programs are each in different colleges at RIT, which added to the challenge.

2 HOW WE STRUCTURED THE COURSE

The project was led by two faculty members, Jesse O'Brien from IGM, and Craig Foster from MI. Jesse and Craig each taught a production-based elective class over two semesters for their respective majors. An attempt was made to merge the two classes into one class, but due to the complexities of course structures between colleges at RIT, we were unable to do that. So we each had our separate classes, and we synchronized our course schedules (no easy feat), and all met as one combined class in a large computer lab.

We ran the two combined classes like a development studio and incorporated Agile development methodologies such as Scrum and standups. We used Slack for online communication and GitHub for source control. Jesse and Craig took on the roles of Studio Heads, Project Managers, Lead Designers, Producers, Product Researchers, and Asset Implementation Experts. The MI students created the 2D and 3D assets, and some of the UI design with Maya and Photoshop. The IGM students did the software programming, in-engine asset implementation, UX design, and UI design. They used Unity, C#, Maya and Photoshop.

This approach had many benefits. The cross-disciplinary collaboration gave our team a broad range of skill sets, and we were able to create a much more complex project. This also gave students the opportunity to work with other students in different disciplines, in a way that is essential in industry but is not typically done in the classroom context.

3 WHAT WE CREATED

We created an anatomically correct interactive arm model in a virtual lab environment. We developed the functionality for the user to select individual muscles and bones, which when selected, activate pop-ups that give information about the selected anatomy. The user can also select and pose the arm, rotating at the shoulder, elbow, and wrist. Muscle visibility can be toggled on or off, to see the skeleton underneath. An interactive light can be selected and moved to see how the anatomy looks under different lighting conditions. The user can write in 3D, move about the environment, and pick up almost everything in the environment and interact with it. We created a tutorial to onboard new users and acquaint them with the controls. We developed VRm with full VR functionality, we ran it on the Oculus Rift, HTC VIVE, Acer Windows Mixed Reality Headset, and the Oculus Quest 2.

4 WHAT WE LEARNED

We took two classes from different majors comprised of students with vastly different skillsets and perspectives from separate colleges at RIT, and we learned how to organize and manage them

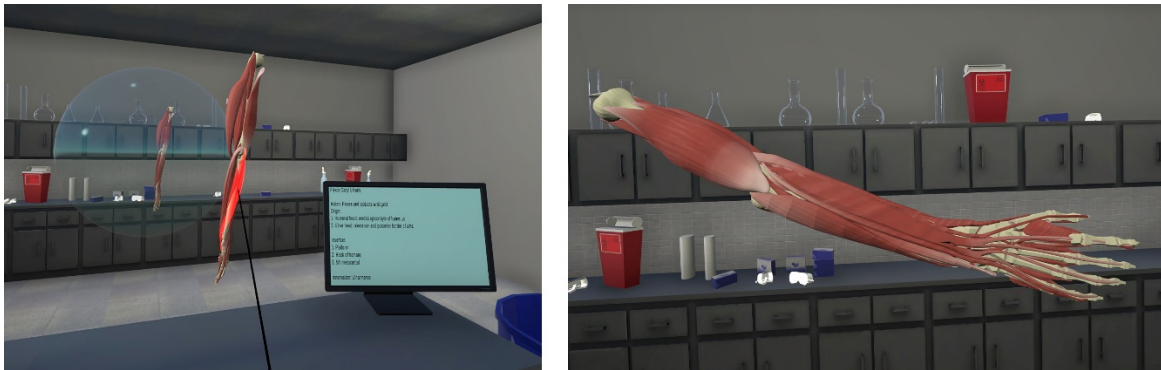


Figure 1: Screenshots of Vrm.

as one class unit. We learned creative ways to navigate the limitations of working with Unity and real-time rendering. These are approaches that are ubiquitous in video game development, but that need to be applied differently in the context of real-time medical simulation for VR. For example, muscle deformations for accurate anatomy are orders of magnitude more complex than typical game mesh deformations.

We learned that creating realistic, selectable, and poseable anatomy is complex in VR. One of our biggest challenges was working out the code and controls for the joint transforms. We tried

three completely different approaches, each more on target than the last. However, we have not yet solved this complex problem fully to our liking.

Virtual reality can be hard for new users. We learned that many people in the medical community are not familiar with VR technology, and our simulation required a carefully designed interface and controls to make the tools as intuitive as possible. We experimented with various control schemes and tutorial methods and learned more throughout that process.